Hindi + English

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Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024-2025)

NEET (UG)
MAJOR TEST # 03
16-02-2025

PRE-MEDICAL : LEADER TEST SERIES / JOINT PACKAGE COURSE

12th Undergoing/Pass Students

Test Type : Full Syllabus

*This Booklet contains **48** pages. इस पुस्तिका में **48** पृष्ठ हैं।*

इस परीक्षा पुस्तिका को तब तक ना खोलें जब तक कहा न जाए।

Do not open this Test Booklet until you are asked to do so.

इस परीक्षा पुस्तिका के पिछले आवरण पर दिए निर्देशों को ध्यान से पढ़ें।

Read carefully the Instructions on the Back Cover of this Test Booklet.

महत्वपूर्ण निर्देश :

1. उत्तर पत्र के **पृष्ठ-1** एवं **पृष्ठ-2** पर ध्यानपूर्वक केवल **नीले/काले** बॉल पॉइंट पेन से विवरण भरें।
2. परीक्षा की अवधि **3 घंटे 20 मिनट** है एवं परीक्षा पुस्तिका में **200** प्रश्न है। प्रत्येक प्रश्न **4** अंक का है। प्रत्येक सही उत्तर के लिए परीक्षार्थी को **4** अंक दिए जाएंगें। प्रत्येक गलत उत्तर के लिए कुल योग में से **एक अंक** घटाया जाएगा। अधिकतम अंक **720** है।
3. इस प्रश्न पत्र के प्रत्येक विषय में **2 खण्ड** है। **खण्ड A** में **35** प्रश्न है (सभी प्रश्न अनिवार्य है) तथा **खण्ड B** में **15** प्रश्न है। परीक्षार्थी इन 15 प्रश्नों में से कोई भी 10 प्रश्न कर सकता है। यदि परीक्षार्थी 10 से अधिक प्रश्न का उत्तर देता है तो हल किये हुए प्रथम 10 प्रश्न ही मान्य होंगे।
4. यदि किसी प्रश्न में एक से अधिक विकल्प सही हो, तो सबसे उचित विकल्प को ही उत्तर माना जायेगा।
5. इस पृष्ठ पर विवरण अंकित करने एवं उत्तर पत्र पर निशान लगाने के लिए केवल **नीले/काले बॉल पॉइंट पेन** का प्रयोग करें।
6. रफ कार्य इस परीक्षा पुस्तिका में निर्धारित स्थान पर ही करें।
7. **परीक्षा सम्पन्ना होने पर, परीक्षार्थी कक्ष/हॉल छोड़ने से पूर्व उत्तर पत्र निरीक्षक को अवश्य सौंप दें। परीक्षार्थी अपने साथ केवल परीक्षा पुस्तिका को ले जा सकते हैं।**
8. परीक्षार्थी सुनिश्चित करें कि इस उत्तर पत्र को मोड़ा न जाए एवं उस पर कोई अन्य निशान न लगाएं। परीक्षार्थी अपना फॉर्म नम्बर प्रश्न पुस्तिका/ उत्तर पत्र में निर्धारित स्थान के अतिरिक्त अन्यत्र न लिखें।
9. उत्तर पत्र पर किसी प्रकार के संशोधन हेतु व्हाइट फ़्लुइड के प्रयोग की अनुमति **नहीं** है।

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1 and Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get 4 marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. In this Test Paper, each subject will consist of **two sections. Section A** will consist of 35 questions (all questions are mandatory) and **Section B** will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
4. In case of more than one option correct in any question, the best correct option will be considered as answer.
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/ Hall. The candidates are allowed to take away this Test Booklet with them.**
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **not** permissible on the Answer Sheet.

किसी भी प्रश्न के अनुवाद में अस्पष्टता के मामले में, अंग्रेजी संस्करण को अंतिम माना जाएगा।

In case of any ambiguity in translation of any question, English version shall be treated as final.

परीक्षार्थी का नाम (बड़े अक्षर में) :

Name of the Candidate (in Capitals) : _____

फॉर्म नम्बर : अंकों में

Form Number : in figures _____

: शब्दों में

: in words _____

परीक्षा केंद्र (बड़े अक्षरों में) :

Centre of Examination (in Capitals) : _____

परीक्षार्थी के हस्ताक्षर :

Candidate's Signature : _____

निरीक्षक के हस्ताक्षर :

Invigilator's Signature : _____

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SECTION - A (PHYSICS)

1. In a circular motion of a particle, the tangential acceleration of the particle is given by $a_t = 12\ ms^{-2}$. The radius of the circle is 3 m. The particle was initially at rest. Time after which total acceleration of the particle makes an angle 45° with the radial acceleration is

(1) 2 s (2) (1/2) s (3) 3 s (4) (1/3) s

2. A satellite of mass 'm' is orbiting the earth (of radius R) at a height equal to radius of earth. The total energy of the satellite in terms of g (acceleration due to gravity on earth's surface) is :-

(1) $-\frac{mgR}{3}$ (2) $-\frac{mgR}{4}$ (3) $-\frac{mgR}{2}$ (4) $-\frac{mgR}{5}$

3. The angular speed of rotation of earth about its axis, at which the weight of man standing on the equator becomes half of his weight at the poles, is given by:

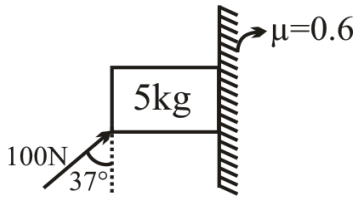
(1) $0.034\ rad\ s^{-1}$

(2) $8.75 \times 10^{-4}\ rad\ s^{-1}$

(3) $1.23 \times 10^{-2}\ rad\ s^{-1}$

(4) $7.65 \times 10^{-7}\ rad\ s^{-1}$

4. A block is in contact with vertical wall as shown. Find friction on block.



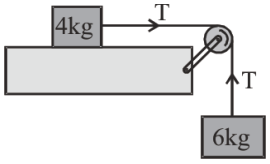
(1) 30 N (2) 36 N

(3) Zero (4) None of these

5. Liquid rises to a height 2 cm in a capillary tube; in that case, the angle of contact between the solid and the liquid is zero. The tube is lowered more now, so that capillary is only 1 cm above the liquid, in this case, the angle of contact between the solid and liquid is :-

(1) 0° (2) 30° (3) 60° (4) 90°

6. Two bodies of mass 6 kg and 4 kg are tied to a string as shown in the adjoining figure. If the table and pulley are frictionless, then acceleration of mass 6 kg will be :
$(g = 10\ ms^{-2})$



(1) $60\ ms^{-2}$ (2) $40\ ms^{-2}$

(3) $6\ ms^{-2}$ (4) $4\ ms^{-2}$

7. A disc and a ring of same mass are rolling and if their kinetic energies are equal, then the ratio of their velocities will be

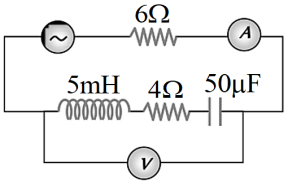
(1) $\sqrt{4} : \sqrt{3}$ (2) $\sqrt{3} : \sqrt{4}$

(3) $\sqrt{3} : \sqrt{2}$ (4) $\sqrt{2} : \sqrt{3}$

8. Thickness of a pencil measured by a screw gauge (least count 0.001 cm) comes out to be 0.802 cm. The percentage error in the measurement is :-

(1) 12.5 % (2) 1.25 % (3) 0.125%(4) 2.125%

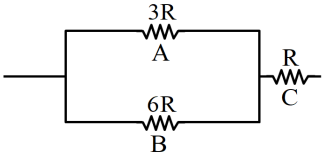
9. In the circuit shown in the figure, the ac source gives a voltage V= 20 cos(2000t). Neglecting source resistance, the voltmeter and ammeter reading will be :



(1) 0 V, 0.47 A (2) 1.68 V, 0.47 A

(3) 0 V, 1.4 A (4) 5.6 V, 1.4 A

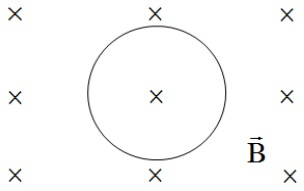
10. Three resistances A, B and C having values 3R, 6R and R respectively forms a network as shown. When some potential differences is applied across the network; the thermal powers dissipated by A, B and C are in the ratio :



(1) 2 : 3 : 4 (2) 2 : 4 : 3

(3) 4 : 2 :3 (4) 3 : 2 : 4

11. A conducting loop is kept perpendicularly in the increasing magnetic field as shown in figure the direction of induced current in the loop will be :-



(1) clockwise

(2) anti-clock wise

(3) zero

(4) insufficient information

12. Light having an energy flux of 40 Wcm^{-2} falls on non-reflecting surface at normal incidence. If the surface has an area of 20 cm^2, the total momentum delivered (for complete absorption) during 10 min is :

(1) 24×10^{-5} kg ms^{-1}

(2) 16×10^{-4} kg ms^{-1}

(3) 102×10^{4} kg ms^{-1}

(4) 1.03×10^{7} kg ms^{-1}

13. A charge (−q) and another charge (+Q) are kept at two points A and B respectively. Keeping the charge (+Q) fixed at B, the charge (−q) at A is moved to another point C such that ABC forms an equilateral triangle of side ℓ. The net work done in moving the charge (−q) is :

(1) $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{\ell}$ (2) $\frac{1}{4\pi\varepsilon_0}\frac{Qq}{\ell^2}$

(3) $\frac{1}{4\pi\varepsilon_0}Qq\ell$ (4) Zero

14. The terminal velocity of a sphere of radius 3 mm falling in a viscous liquid is 10 cms^{-1}, then the terminal velocity of ball of radius 6 mm of same material falling in same liquid will be :-

(1) 10 cms^{-1} (2) 20 cms^{-1}

(3) 40 cms^{-1} (4) 50 cms^{-1}

15. In gaps of meter bridge two unknown resistances A and B are connected. Null point is obtained when the wire gets divided in ratio of 3 : 4. When each of resistance is increased by 40Ω, the null point, divides the wire in ratio 7 : 8. The values of resistance A and B are respectively

(1) 20Ω; 30Ω (2) 30Ω, 40Ω

(3) 40Ω, 60Ω (4) 60Ω, 80Ω

16. A coin is dropped in a lift. It takes time t_1 to reach the floor when lift is stationary. It takes time t_2 when lift is moving up with constant acceleration, then:

(1) $t_1 = t_2$ (2) $t_1 > t_2$

(3) $t_2 > t_1$ (4) $t_1 >> t_2$

17. A ball thrown by one player reaches to other player in 2 sec. The maximum height attained by the ball above the point of projection will be :

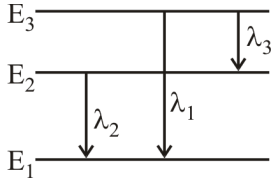
(1) 2.5m (2) 5m (3) 7.5m (4) 10m

18. A bulb lamp emits light of mean wavelength of 4500 Å. The lamp is rated at 150 W and 8% of the energy appears as emitted light. How many photons are emitted by the lamp per second ?

(1) 2.717×10^{18} (2) 27.17×10^{18}

(3) 3×10^{20} (4) 3×10^{15}

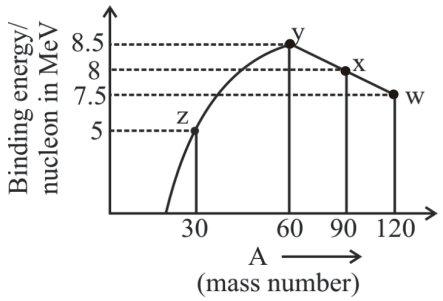
19. Transition between three energy levels in a particular atom give rise to three spectral lines of wavelength, in increasing magnitudes, λ_1, λ_2 and λ_3. Which one of the following equation correctly relates λ_1, λ_2 & λ_3 ?



(1) $\lambda_1 = \lambda_2 - \lambda_3$ (2) $\lambda_1 = \lambda_3 - \lambda_2$

(3) $\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$ (4) $\frac{1}{\lambda_1} = \frac{1}{\lambda_3} - \frac{1}{\lambda_2}$

20. Binding energy per nucleon v/s mass number curve for nuclei is shown in the figure. w, x, y and z are four nuclei indicated on the curve. The process that would release energy is :-



(1) $y \longrightarrow 2z$ (2) $w \longrightarrow x + z$

(3) $w \longrightarrow 2y$ (4) $x \longrightarrow y + z$

21. A nucleus with z = 92 emits the following in a sequence : α, α, β^-, β^-, α, α, α, α, β^-, β^-, α, β^+, β^+ and α. The atomic number of the resulting nucleus is -

(1) 76 (2) 78 (3) 80 (4) 72

22. What will be the height of image, when an object of height 2 mm is placed on the axis of a convex mirror (of radius of curvature 40 cm) at a distance 20 cm from mirror.

(1) 20 mm (2) 10 mm

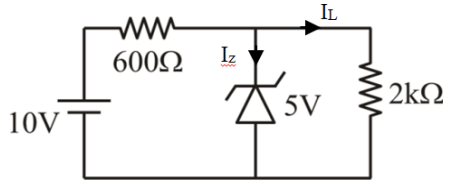
(3) 6 mm (4) 1 mm

23. The length of a telescope is 36 cm. The focal lengths of its lenses can be

(1) 30 cm, 6 cm (2) –30 cm, –6 cm

(3) –30 cm, 6 cm (4) 36 cm, 0 cm

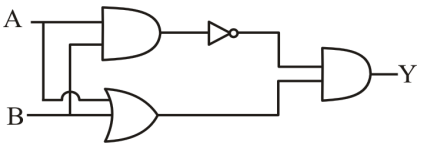
24. In the given circuit, the value of I_L (in mA) will be



(1) 2.5 (2) 1.5

(3) 3.5 (4) 4.5

25. Which one of the following will be the output of the given circuit ?



(1) $\overline{A + B}$ (2) $\overline{A.B}$

(3) A.B (4) $\bar{A}B + A\bar{B}$

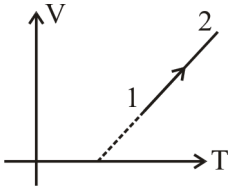
26. A monoatomic ideal gas undergoes an adiabatic process at room temperature. The relation between temperature and volume is "TV^x = constant", then x is

(1) $\frac{2}{5}$ (2) $\frac{2}{3}$ (3) $\frac{5}{3}$ (4) $\frac{3}{5}$

27. A lead bullet of 20g travelling at 300 m/sec stikes against a block of wood and comes to rest. Assuming 50% of heat is absorbed by the bullet, the increase in its temperature is -
(take specific heat of lead = 150 J/kg–K)

(1) 100°C (2) 150°C (3) 125°C (4) 200°C

28. A graph between volume V and absolute temperature T was obtained when a given mass of a gas is heated. During the heating process from state-1 to state-2, the pressure



(1) remained same (2) decreased

(3) increased (4) none of above

29. The dimensional formula for magnetic moment is :

(1) $[M^0L^2T^0A^1]$ (2) $[M^0L^2T^0A^2]$

(3) $[M^0L^1T^0A^2]$ (4) $[M^0L^0T^1A^1]$

30. If a pipe gives notes of frequencies 375 Hz, 625 Hz and 875 Hz, what is fundamental frequency of the pipe and its type ?

(1) 250 Hz, open (2) 125 Hz, open

(3) 250 Hz, close (4) 125 Hz, close

31. Two sitar strings A and B playing the note "Sa" are slightly out of tune and produce beats of frequency 5 Hz. The tension of the string B is slightly decreased and the beat frequency is found to decrease to 3 Hz. What is the original frequency of B, if frequency of A is 255 Hz?

(1) 250 Hz (2) 258 Hz

(3) 260 Hz (4) 252 Hz

32. In a Young's double slit experiment, the fringe width with a light of 4800Å is found to be 0.03 mm. What will be the fringe-width if the light of wavelength 6400Å is used -

(1) 0.01 mm

(2) 0.02 mm

(3) 0.03 mm

(4) 0.04 mm

33. In fraunhofer diffraction, the light of wavelength 6328 Å is incident on a slit of width 0.2 mm perpendicularly, then angular fringe width of secondary maxima or minima will be -

(1) 0.36^o (2) 0.18^o

(3) 0.72^o (4) 0.09^o

34. A rigid body of mass 4 kg initially at rest moves under the action of an applied horizontal force of 18 N on a table with coefficient of kinetic friction 0.2. Work done by the applied force on the body in 10 s will be

(1) 1250 J (2) 2250 J

(3) 2500 J (4) 1000 J

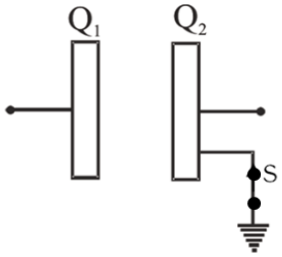
35. A straight horizontal long wire carries 30 A current. The linear mass density of wire is 45 g/m. What is magnitude of magnetic field needed to balance the wire in air?

(1) 15×10^{-3} T (2) 15 T

(3) 15×10^{-4} T (4) 15×10^{-2} T

SECTION - B (PHYSICS)

36. The two plates of a parallel plate capacitor are given charges Q_1 and Q_2 .The capacity of the capacitor is C. When the switch (S) is closed, mark the correct statement. (Assume both Q_1 and Q_2 to be positive)



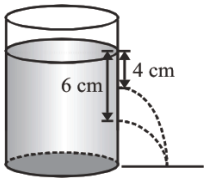
(1) The charge flown through switch is zero.

(2) The charge flown through switch is $\frac{Q_1 + Q_2}{2}$

(3) Potential difference across the capacitor plates is $\frac{Q_1}{C}$.

(4) The charge on the capacitor is $\frac{Q_1}{2}$

37. Figure shows two holes in a wide tank containing a liquid column. The water streams coming out of these holes strike the ground at the same point. The height of liquid column in the tank is :-



(1) 10 cm (2) 8 cm

(3) 9.8 cm (4) 980 cm

38. ***Assertion*** **:-** When no external torque is applied and moment of inertia of a rotating body changes, then its angular momentum remains conserved, but its kinetic energy changes.
Reason **:-** Angular momentum does not depend upon moment of inertia of the body.

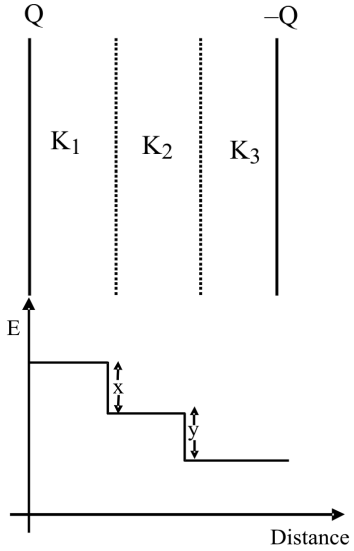
(1) **(A)** is correct but **(R)** is not correct

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

39. Electric field variation between the plates of capacitor is shown. Find relation between K_1, K_2, K_3, if $2x = y$.



(1) $\frac{1}{K_2} = \frac{1}{K_1} + \frac{1}{K_3}$

(2) $\frac{3}{K_2} = \frac{2}{K_1} + \frac{1}{K_3}$

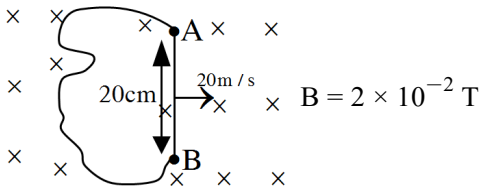
(3) $\frac{2}{K_2} = \frac{1}{K_1} + \frac{4}{K_3}$

(4) None

40. A moving ball collides head on a similar ball at rest. If $\frac{1}{4}$th of kinetic energy is lost by the impact, the value of coefficient of restitution is :-

(1) $\frac{1}{2\sqrt{2}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{\sqrt{3}}{2}$

41. A closed loop is moving with constant velocity in uniform magnetic field as shown. Potential difference and current in element AB will be



(1) 0.08 volt, 0.1 A (2) 0.04 volt, 0.02 A

(3) 0.08 volt, zero (4) zero, zero

42. Two point charges (+Q) and (−2Q) are fixed on the X-axis at positions a and 2a from origin respectively. At what position on the axis, the resultant electric field is zero ?

(1) Only $x = \sqrt{2}a$ (2) Only $x = -\sqrt{2}a$

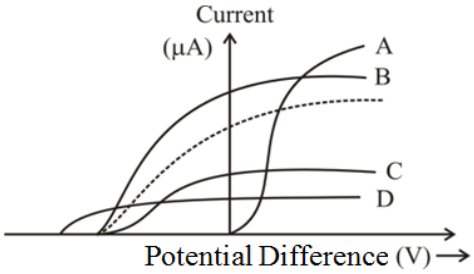
(3) Both $x = \pm\sqrt{2}a$ (4) Only $x = \frac{3a}{2}$

43. A proton is accelerated through a potential difference of 100 volt. To have same de-broglie wavelength, what potential difference must be applied across doubly ionised $^{8}_{4}Be$?

(1) 1.25 volt (2) 12.5 volt

(3) 0.625 volt (4) 6.25 volt

44. A photoelectric cell is connected to a source of variable potential difference and the resulting photoelectric current (μA) is plotted against the applied potential difference (V). The graph in the broken line represents one situation for a given frequency and intensity of the incident radiation. If the frequency is increased and intensity is reduced, which of the following graphs of unbroken line represents the new situation?



(1) A (2) B

(3) C (4) D

45. A linear harmonic oscillator has a total mechanical energy 300 J. If its potential energy at mean position is 100 J, then find kinetic energy at $x = +\frac{A}{\sqrt{2}}$.

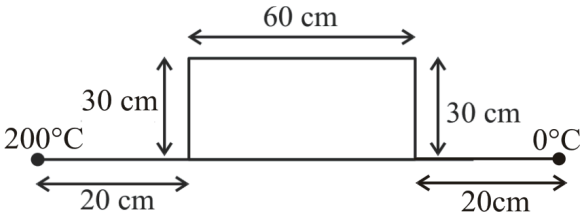
(1) 100 J (2) 200 J

(3) 300 J (4) 400 J

46. Angle of minimum deviation for a prism of refractive index 1.5 is equal to the angle of prism. The angle of prism is (cos41° = 0.75)

(1) 62° (2) 41° (3) 82° (4) 31°

47. Consider the situation as shown in figure. Find the amount of heat flowing per second through a cross-section of the bent part, if the total heat taken out per second from the end at 200°C is 150 J.



(1) 130 J (2) 50 J

(3) 70 J (4) 80 J

48. A man travels 30 m along $3\hat{i} + 4\hat{j}$ and then moves 'd' meters perpendicular to initial direction so that his total displacement is along x-axis. What is the value of 'd' ?

(1) 24m (2) 36m

(3) 16m (4) 40m

49. A long straight wire, of radius a, carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire, at distance $\frac{a}{3}$ and 2a respectively from the axis of the wire, is :-

(1) $\frac{3}{2}$ (2) 2 (3) $\frac{2}{3}$ (4) $\frac{1}{2}$

50. Which of the following relation is incorrect ?

(a) $B = \mu_0 (H + I)$ (b) $B = \mu_0 (H + \chi_m)$

(c) $\mu_r = 1 + \chi_m$ (d) $\chi_m = I/H$

(Here, I = intensity of magnetisation and H = magnetising field)

(1) only a

(2) only b

(3) a, c, d

(4) All of above

SECTION - A (CHEMISTRY)

51. Find the formula of halide of a metal whose successive ionization energies are 10, 16, 25, 400, 500 eV mol^{-1} respectively :-

(1) MX (2) MX_2 (3) MX_3 (4) M_2X

52. Which of the following electronic configuration will have nearly zero electronegativity ?

(1) $1s^2\ 2s^2\ 2p^5$ (2) $1s^2\ 2s^2\ 2p^3$

(3) $1s^2$ (4) $1s^2\ 2s^2$

53. The shape of 'AsF_5' according to V.S.E.P.R theory :-

(1) Pentagonal Pyramidal

(2) Trigonal bipyramidal

(3) Square Pyramidal

(4) Octahedral

54. Solid PCl_5 exist as :-

(1) $[PCl_4]^+$ & $[PCl_6]^-$

(2) PCl_3 & Cl_2

(3) $[PCl_4]^-$ & $[PCl_6]^+$

(4) $[PCl_3]^+$ & $[PCl_3]^-$

55. Which of the following statement regarding NH_3 and NF_3 is correct :

(1) NH_3 has pyramidal and NF_3 has trigonal planar shape

(2) Bond angle in NH_3 is smaller than NF_3

(3) Resultant dipole moment of NH_3 is $(4.90 \times 10^{-30}$ cm) and that of NF_3 is $(0.8 \times 10^{-30}$ cm)

(4) They both are sp^2 hybridised

56. Stability of the species Li_2, Li_2^- and Li_2^+ increases in the order of :-

(1) $Li_2 < Li_2^+ < Li_2^-$ (2) $Li_2^- < Li_2^+ < Li_2$

(3) $Li_2 < Li_2^- < Li_2^+$ (4) $Li_2^- < Li_2 < Li_2^+$

57. Match the coordination number and type of hybridisation with distribution of hybrid orbitals in space based on Valence bond theory.

	Coordination number and type of hybridisation		Distribution of hybrid orbitals in space
(a)	4, sp^3	(i)	trigonal bipyramidal
(b)	3, sp^2	(ii)	octahedral
(c)	5, sp^3d	(iii)	tetrahedral
(d)	6, sp^3d^2	(iv)	trigonal planar

Select the correct option :

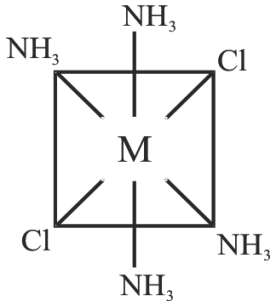
(1) (a)-(ii) (b)-(iii) (c)-(iv) (d)-(i)

(2) (a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)

(3) (a)-(iv) (b)-(i) (c)-(ii) (d)-(iii)

(4) (a)-(iii) (b)-(i) (c)-(iv) (d)-(ii)

58. The isomer can be marked as :-



(1) Cis isomer (2) Trans isomer

(3) d-isomer (4) ℓ-isomer

59. Which among the following statements is incorrect.

(1) In d-block elements consecutive oxidation state differ by unity.

(2) p-block element form's only acidic oxide

(3) In a group of p-block lower oxidation states are favoured by the heavier members.

(4) In a group of d-block lower oxidation states are favoured by the lighter member.

60. The most basic oxide of lanthanoids is :-

(1) $Pr(OH)_3$ (2) $Eu(OH)_3$

(3) $Gd(OH)_3$ (4) $Lu(OH)_3$

61. Which of the statements is not true?

(1) $K_2Cr_2O_7$ solution in acidic medium is orange

(2) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7

(3) On passing H_2S through acidified $K_2Cr_2O_7$ solution, a milky colour is observed

(4) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis

62. If the fertilizers listed below are priced according to their nitrogen content, which will be the least expensive per 50 kg bag ?

(1) Urea, $(NH_2)_2CO$

(2) Ammonia, NH_3

(3) Ammonium nitrate, NH_4NO_3

(4) Guanidine, $HNC(NH_2)_2$

63. The two electrons X and Y have following sets of quantum numbers -

	n	ℓ	m	s
X	3	2	–2	+1/2
Y	3	0	0	+1/2

which of the following is the correct statement ?

(1) X and Y have same energy

(2) X has greater energy than Y

(3) X has less energy than Y

(4) X and Y represent same electron

64. The reaction for which $K_c = 2.3 \times 10^3$ moles/litre is -

(1) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(3) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

65. 2 moles each of A and B were taken in a container and following reaction take place $2A(g) + B(g) \rightleftharpoons 2C(g) + D(g)$. When the system attained equilibrium then which is true :-

(1) [A] > [B]

(2) [A] < [B]

(3) [A] = [B]

(4) Can't predict

66. **Statement-I :-** pH of pure water increases with increase in temperature.
Statement-II :- Degree of ionisation increases with increases in temperature.

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

67. For the reaction $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$; $K_c = 4$. If equilibrium concentration of $A_{(g)}$ and $B_{(g)}$ are found to be 0.1 M and 0.4 M respectively. Determine equilibrium concentration of $C_{(g)}$.

(1) 0.4 M

(2) 0.3 M

(3) 0.2 M

(4) 0.1 M

68. **Assertion :-** Vapour pressure of a liquid increases on increasing surface area.
Reason :- Rate of vaporisation of a liquid is directly proportional to surface area.

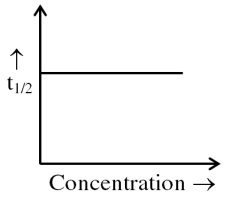
(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

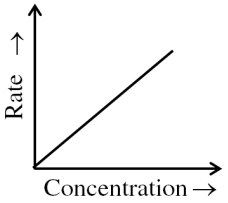
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

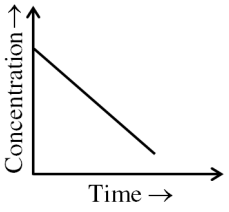
(3) **Assertion** is true but **Reason** is false.

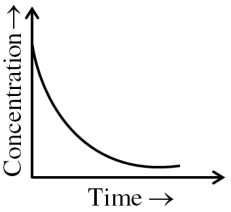
(4) **Assertion** is false but **Reason** is true.

69. Which of the following is incorrect for first order reaction ?
$A \longrightarrow B$

(1) 

(2) 

(3) 

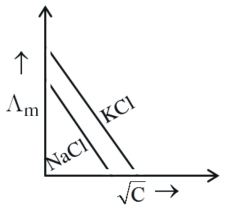
(4) 

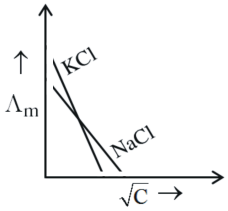
70. For a first order reaction $A \longrightarrow P$, $t_{1/2}$ (Half life) is 10 days. The time required for $1/4^{th}$ conversion of A (in days) is -

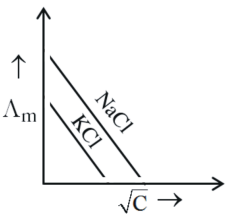
(1) 5 (2) 3.2

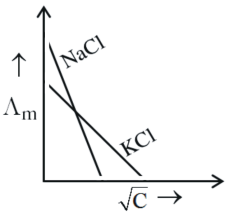
(3) 4.1 (4) 2.5

71. Which one of the following graphs between molar conductivity (Λ_m) versus $\sqrt{C}$ is correct ?

(1) 

(2) 

(3) 

(4) 

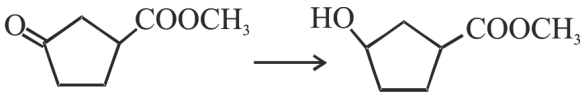
72. Electrode potential of the half cell
$Hg(\ell) \mid Hg_2Cl_2(s) \mid Cl^-(aq)$ can be increased by

(1) Increasing $[Cl^-]$

(2) Decreasing $[Cl^-]$

(3) Increasing $Hg_2Cl_2(s)$

(4) Decreasing $Hg(\ell)$

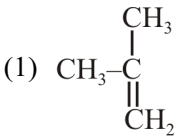
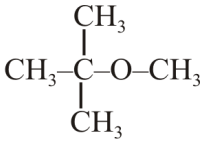
73. 

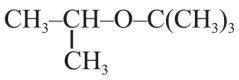
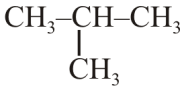
Which of the following reagents should be used to carry out the above conversion :-

(1) $LiAlH_4$ (2) $NaBH_4$

(3) Ni, H_2 (4) Zn–Hg/HCl

74. $CH_3-\underset{\underset{CH_3}{|}}{CH}-ONa + (CH_3)_3C-Cl \xrightarrow{\Delta}$ Major product

(1)  (2) 

(3)  (4) 

75. $CH_3-CH=CH-CH=CH_2 \xrightarrow[Zn]{O_3/H_2O}$

Which will not form ?

(1) $CH_3-CH=O$ (2) $CH_3-CH_2-CH=O$

(3)  (4) $CH_2=O$

76. Which of the following match is wrong :-

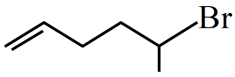
(1) Vitamin B_6 – Pyridoxine

(2) A sequence of Base – DNA finger printing

(3) Amino acid –Zwitter ion

(4) All match are correct

77. IUPAC name of given compound is :-



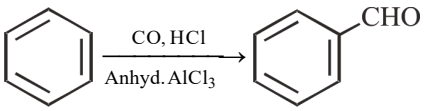
(1) 5-Bromo-5-Methyl pent-1-ene

(2) 5-Bromo hex-1-ene

(3) 2-Bromo hex-5-ene

(4) 1-Bromo-1-Methyl pent-4-ene

78. What will be the name of the given reaction :



(1) Gatterman Koch reaction

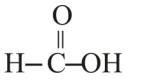
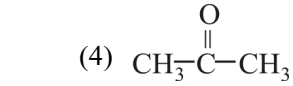
(2) Etard reaction

(3) Gattermann reaction

(4) Rosenmund reaction

79. Which of the following doesn't reduce Fehling's solution?

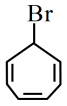
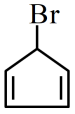
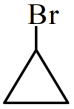
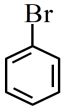
(1) CH_3-CHO (2) $H-CHO$

(3)  (4) 

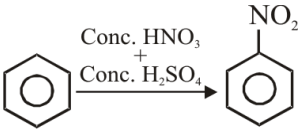
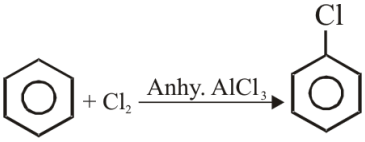
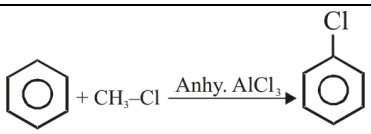
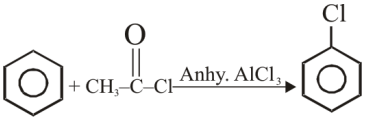
80. Total number of structure isomeric alcohol with molecular formula $C_4H_{10}O$ is :-

(1) 3 (2) 2 (3) 4 (4) 6

81. Which among the following will undergo hydrolysis rapidly :-

(1)  (2)  (3)  (4) 

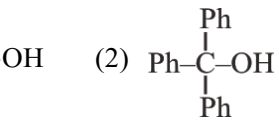
82. Match the following :

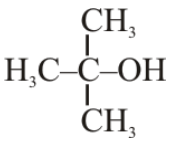
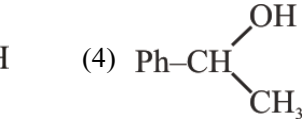
Column-I		Column-II	
(i)		(A)	Halogenation
(ii)		(B)	Nitration
(iii)		(C)	Friedal–crafts acylation
(iv)		(D)	Friedal–crafts alkylation

Options :

	i	ii	iii	iv
(1)	A	B	C	D
(2)	B	A	C	D
(3)	B	A	D	C
(4)	A	C	D	B

83. Which of the following alcohols will not react with Cu/Δ ?

(1) CH_3-CH_2-OH (2) 

(3)  (4) 

84. In carius method of estimation of halogen 0.15 gm of an organic compound gave 0.12 gm of AgBr. Find out percentage of bromine in the compound (Molar mass of AgBr : 188 gm)

(1) 10.74 %

(2) 21.28 %

(3) 42.55 %

(4) 34.04 %

85. n-Butane, $C_2H_5-O-C_2H_5$,
(a) (b)

$CH_3-(CH_2)_3-OH$, CH_3-CH_3
(c) (d)

Correct order of boiling point is :-

(1) c > b > a > d (2) b > c > a > d

(3) c > a > b > d (4) a > b > c > d

SECTION - B (CHEMISTRY)

86. Which of the following order is incorrect :-

(1) $BF_3 < PF_3$ Dipole moment

(2) N–N < P–P bond strength

(3) $PH_3 > NH_3$ solubility in water.

(4) $Ne < Ne^+$ (I.P)

87. Considering ethene molecule, which of the following statements is/are correct ?

I. C–C bond is formed by overlapping of sp^2-sp^2 hybrid orbitals.

II. C–H bond is formed by overlapping of sp^2–s orbitals.

III. $p\pi-p\pi$ bond is present.

IV. H–C–H bond angle is greater than H–C–C bond angle.

V. Nodal plane of Pi bond and molecular plane are same.

(1) I, II, III (2) I, II, III, IV

(3) I, II, III, IV, V (4) I, II, III, V

88. Which of the following orders is correct :

(1) Thermal stability : $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$

(2) Basic Nature : ZnO > BeO < MgO > CaO

(3) Solubility in water : LiOH > NaOH > KOH > RbOH > CsOH

(4) Melting point : NaCl > RbCl > CsCl > KCl

89. ***Assertion (A)*** :- Fe^{+3} not used in brown ring test of NO_3^{-1}.

Reason (R) :- Reduction of NO_3^- takes place during test.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

90. Which of the following statement is not correct?

(1) The stability constant of $[Co(NH_3)_6]^{3+}$ is greater than that of $[Co(NH_3)_6]^{2+}$.

(2) Among F^-, Cl^-, Br^- and I^-, F^- forms strongest complexes due to small size.

(3) $[Cu(NH_3)_4]^{2+}$ is thermodynamically more stable than $[Zn(NH_3)_4]^{2+}$.

(4) Among $[Fe(CN)_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$ and $[Fe(en)_3]^{3+}$, $[Fe(CN)_6]^{3-}$ is least stable.

91. In which of the following conversion Blue colour is appeared.

(1) $MnO_4^- + SO_2 \rightarrow$

(2) $Cr_2O_7^{-2} + KI \rightarrow$

(3) $K_2Cr_2O_7 + NaCl + \text{conc. } H_2SO_4 \rightarrow$

(4) $CuSO_4 + KI + \text{Starch} \rightarrow$

92. Calorific value of hydrogen gas is -143 kJ g^{-1}. The standard enthalpy of formation of H_2O will be -

(1) -143 kJ mol^{-1} (2) $+143$ kJ mol^{-1}

(3) -286 kJ mol^{-1} (4) $+286$ kJ mol^{-1}

93. The spontaneous nature of reaction is impossible if

(1) ΔH is +ve ; ΔS is also positive

(2) ΔH is $-$ve ; ΔS is also negative

(3) ΔH is $-$ve ; ΔS is positive

(4) ΔH is +ve ; ΔS is negative

94. The dissociation constant values of three acids A, B and C are 1.8×10^{-5}, 1.5×10^{-4} and 5.2×10^{-8} respectively. Correct order of increasing basic nature of their conjugate bases is -

(1) A > B > C (2) B > A > C

(3) B < A < C (4) A < B < C

95. A sulphur-containing species that can not act as reducing agent is

(1) SO_2 (2) SO_3^{-2} (3) H_2SO_4 (4) S^{-2}

96. Which of the following statements regarding Henry's law is incorrect ?

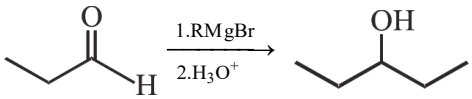
(1) Higher the value of K_H at a given pressure, higher is the solubility of gas in liquid.

(2) Different gases have different K_H (Henry's law constant) values at the same temperature.

(3) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.

(4) The value of k_H increases with increase of temperature and k_H is function of the nature of the gas.

97.

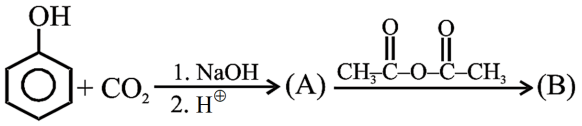


R is :-

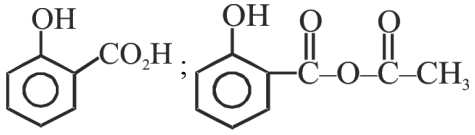
(1) CH_3- (2) CH_3-CH_2-

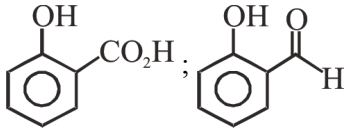
(3) $CH_3CH(CH_3)-$ (4) $CH_3-CH_2-CH_2-$

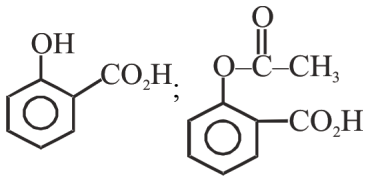
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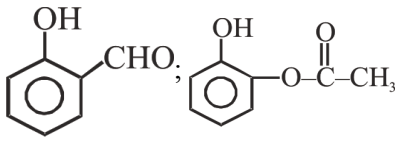


Product (A) and (B) respectively are :-

(1) 

(2) 

(3) 

(4) 

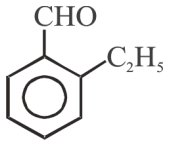
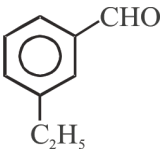
99. An aromatic compound 'X' with molecular formula $C_9H_{10}O$ gives the following chemical tests :

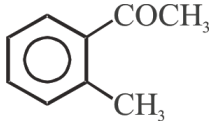
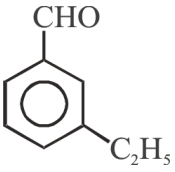
(i) show 2, 4-DNP reaction

(ii) reduces Tollens' reagent

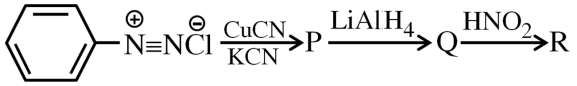
(iii) undergoes Cannizzaro reaction and

(iv) on vigorous oxidation, 1,2-benzenedicarboxylic acid is obtained X is :

(1)  (2) 

(3)  (4) 

100. Identify R in the following sequence of reaction –



(1) Benzoic acid

(2) Phenyl acetic acid

(3) Benzyl alcohol

(4) Benzamide

SECTION - A (BOTANY)

101. Perisperm is :-

(1) Persistent nucleus in seed

(2) Ovary wall

(3) Ovule coat

(4) Persistent nucellus in seed

102. Which of the following can withstand high temperatures, strong acids and alkali ?

(1) Pecto-cellulose

(2) Sporopollenin

(3) Intine

(4) Cutin

103. In Mendelian experiment which of the following trait is dominant -

(1) Yellow pod colour (2) Yellow seed colour

(3) Wrinkled seed (4) Dwarf plant

104. Following given table is regarding ABO blood group, Identify W, X, Y, Z.

Genotype	Blood group
$I^A\ I^A$	A
W	A
$I^B\ I^B$	B
X	B
$I^A\ I^B$	Y
Z	O

	W	X	Y	Z
(1)	$I^A I^0$	$I^B I^0$	$I^0 I^0$	A B
(2)	$I^A I^0$	$I^B I^0$	A B	$I^0 I^0$
(3)	$I^B I^0$	$I^A I^0$	A B	$I^0 I^0$
(4)	$I^B I^A$	$I^A I^A$	A B	$I^0\ I^0$

105. Read the following statements carefully :

(a) Linkage is physical association of genes on a chromosome.

(b) In drosophila, genes of body colour and eye colour are tightly linked on X-chromosome.

(c) In honey bee a fertilised egg develops as a male (drone).

(d) In grasshopper males have only one X-chromosome besides the autosomes.

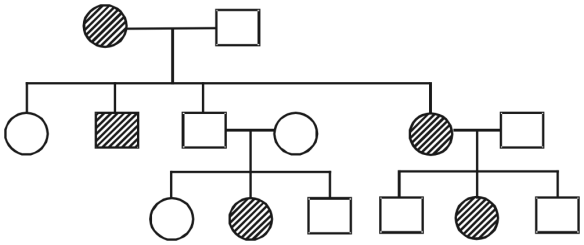
(e) Drosophila can complete their life cycle in about two months.

Choose the correct option having all correct options.

(1) a, b, c and d (2) c and e

(3) a, b and d (4) c, d and e

106. Given pedigree chart show which type of inheritance :-



(1) X-linked recessive

(2) Autosomal dominant

(3) Autosomal recessive

(4) X-linked dominant

107. Read the following statements :-

(i) Eukaryotic mRNA is polycistronic whereas prokaryotic mRNA is monocistronic

(ii) Prokaryotic mRNA is cap-less and tail-less whereas eukaryotic mRNA is with cap and tail

(iii) The precursor of prokaryotic mRNA is hnRNA

(iv) In Replication only one of the strand act as a template.

(v) In eukaryotes, transcription occurs in nucleus and translation in cytoplasm

Which of the above statements are correct ?

(1) (ii), (iii) and (v) (2) (iii), (iv) and (v)

(3) (ii) and (v) (4) (iii) and (iv)

108. Identify the amino acid sequence coded by the given mRNA sequence.
5'AUGUUUUUCGUGAUAUGG–3'

(1) Met–Phe–Leu–Ser–Ile

(2) Met–Phe–Phe–Val–Ile–Trp

(3) Met–Leu–Leu–Ser–Ile–Tyr

(4) Met–Leu–Leu–Val–Cys–Trp

109. During DNA replication one strand is formed continuous and second strand is formed discontinuous because:-

(1) DNA synthesis is bidirectional

(2) DNA replication is semiconservative

(3) DNA chain formation take place in 5' to 3' direction only

(4) DNA replication is semidiscontinuous

110. Sequencing the whole set of genome that contains all the coding and noncoding parts is :-

(1) Expressed sequence tags

(2) Sequence Annotation

(3) Bioinformatics

(4) Electrophoresis

111. The total annual net productivity of whole biosphere is :

(1) 70 billion ton (2) 7 million ton

(3) 170 billion ton (4) 70 million ton

112. Which one is the correct food chain?

(1) Eagle ⟶ Snake ⟶ Grasshopper ⟶ Grass ⟶ Frog

(2) Frog ⟶ Snake ⟶ Eagle ⟶ Grasshopper ⟶ Grass

(3) Grasshopper ⟶ Grass ⟶ Snake ⟶ Frog ⟶ Eagle

(4) Grass ⟶ Grasshopper ⟶ Frog ⟶ Snake ⟶ Eagle

113. Read the following statements carefully and mark the correct option given below
Statement-I : India is one of the 12 mega diversity countries of the world.
Statement-II : India has only 2.4% of the world's land area.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

114. Match the List-I with-II :-

List-I		List-II	
(A)	Narrowly utilitarian	(i)	Hot-spot
(B)	Broadly utilitarian	(ii)	Seed bank
(C)	In situ conservation	(iii)	Photosynthesis
(D)	Ex-situ conservation	(iv)	Food & Fibre

(1) A-iv, B-iii, C-ii, D-i

(2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-iii, C-i, D-ii

(4) A-ii, B-iii, C-iv, D-i

115. Consider the following four statements and select the option which includes all the correct ones :-
(a) Sparrow eating any seed is no less a predator.
(b) Monarch butterfly is highly tasteful to its predator.
(c) Nearly 25% of all insects are known to be phytophagous.
(d) Thorns are the most common morphological defence mechanism.
Options :-

(1) Statements : (a), (c), (d)

(2) Statements : (a), (b)

(3) Statements : (a), (b), (c)

(4) Statements : (b), (d)

116. The formula for exponential population growth is :-

(1) $dN/dt = rN$

(2) $dt/dN = rN$

(3) $\frac{dN}{N} = rdt$

(4) $rN/dN = dt$

117. **Statement - I :** The interaction where one species is benefitted and other is neither benefitted nor harmed is called commensalism.
Statement - II : Barnacles growing on the back of whale is an example of commensalism.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

118. Each character is given equal importance and at the same time hundred of characters can be considered in.

(1) Numerical taxonomy

(2) Natural classification

(3) Chemotaxonomy

(4) Cytotaxonomy

119. Match the column-I with column-II :-

Column-I		Column-II	
A.	Deuteromycetes	I.	*Rhizopus & mucor*
B.	Phycomycetes	II.	Sac fungi
C.	Basidiomycetes	III.	Bracket fungi
D.	Ascomycetes	IV.	Fungi imperfecti

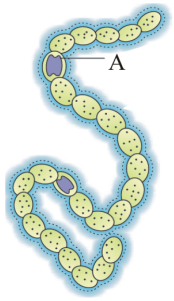
(1) A-III, B-II, C-I, D-IV

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-IV, B-III, C-I, D-II

120. Identify 'A' in the figure given below and to which process it is related ?



(1) Holdfast - Nitrogen fixation

(2) Heterocyst - Nitrogen fixation

(3) Heterocyst - Production of methane

(4) Pyrenoids - Storage bodies

121. (i) *Selaginella* and *Salvinia* are heterosporous plant.
(ii) Gametophyte of pteridophyte is called prothallus.
(iii) In *Cycas*, stems are unbranched.
(iv) Gymnosperms are heterosporous.
How many statement/s is/are correct ?

(1) Two (2) Three (3) One (4) Four

122. Transverse section of a plant organ show vessels but does not show companion cells, This organ may belongs to :

(1) *Yucca* (2) *Ephedra*

(3) *Pinus* (4) *Maize*

123. The endodermis of dicot stem is also referred as –

(1) Bundle sheath

(2) Casparian strip

(3) Conjunctive tissue

(4) Starch sheath

124. Large, empty colourless cells present on upper surface of grass leaf are :-

(1) Palisade parenchyma

(2) Spongy parenchyma

(3) Subsidiary cells

(4) Bulliform cells

125. In leaves, mesophyll is the component of :-

(1) Epidermal tissue system

(2) Ground tissue system

(3) Vascular tissue system

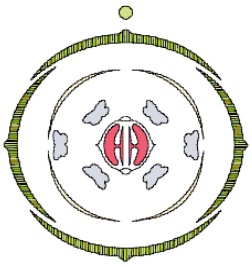
(4) Both (1) & (2)

126. False septum is found in the ovary of flower of :

(1) Sunflower (2) Pea

(3) Primrose (4) Mustard

127. Which of the floral formula is correct for given floral diagram ?



(1) $\oplus \, \delta\!\!\!\!\!\!\;\!\!\!\!\!\!\;\!\!\!\!\!\!\;

135. Which one of the following statements is wrong?

(1) Algin and carrageen are products of algae

(2) Agar-agar is obtained from *Gelidium* and *Gracilaria*

(3) *Chlorella* and *Spirullina* are used by space travellers as a food

(4) Mannitol is stored food in Rhodophyceae

SECTION - B (BOTANY)

136. **Assertion :** The transcription and translation can be coupled in bacteria.
Reason : In bacteria, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

137. How many statements are **correct** ?
(a) Some animal like Cockroaches, Crow etc are omnivores
(b) Detritus food chain (DFC) begins with dead organic matter
(c) Each tropic level has a certain mass of living material at a particular time called as standing crop
(d) Species composition and stratification are two main structural features of an ecosystem

(1) Two (2) Three (3) One (4) All

138. The number of gametes taking part in double fertilization is :-

(1) 3 (2) 4 (3) 2 (4) 5

139. In the inheritance of size of starch grain in *Pisum sativum*, the F_1 had a phenotype that :-

(1) Resemble both of the parents

(2) Resemble none of the parents

(3) Resemble either one of the parents

(4) Both (1) and (3)

140. A normal visioned man whose father was colourblind marries a normal woman whose father was also colourblind. They have one son. What is the percentage of son to be colourblind?

(1) 0% (2) 50% (3) 100% (4) 75%

141. RNA polymerase–I is not responsible for the synthesis of -

(1) 5 s rRNA (2) 28 s rRNA

(3) 5.8 s rRNA (4) 18 s rRNA

142. Chemosynthetic autotrophic bacteria like *Nitrosomonas* obtain, energy from _____ for their ATP production

(1) Oxidation of organic substances

(2) Reduction of organic substances

(3) Oxidation of inorganic substances

(4) Reduction of inorganic substances

143. Pericycle and endodermis are not found in

(1) Dicot stem (2) Dicot root

(3) Monocot stem (4) Monocot root

144. Read the following statements and choose the set of correct statement :-
(A) In same plants like *Monstera* roots arise from parts of the plant other than the radicle.
(B) The main function of the root system are absorption of water and minerals from the soil and providing a proper anchorage to the plant parts.
(C) Leaves orginate from shoot apical meristem and are arranged in an acropetal order.
(D) In *Alstonia,* a single leaf arises at each node of stem
(E) The flower are hypogynous in cucumber.
Choose the correct from the option given below :-

(1) B, D, E only (2) A, C, D only

(3) C, D, E only (4) A, B, C only

145. In mango and coconut, the fruit is known as a

(1) Capsule (2) Berry

(3) Drupe (4) Dry fruit

146. _____ is a cofactor for the proteolytic enzyme carboxypeptidase.

(1) Cn (2) Zn (3) Mg (4) Fe

147. Site of C_3 cycle in C_4 plants is :-

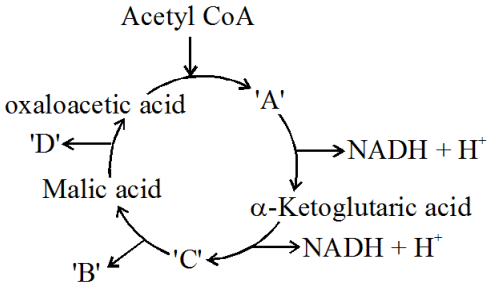
(1) Mesophyll cell (2) Thylakoid membrane

(3) Bundle sheath cell (4) None of these

148. Yeasts poison themselves to death when the concentration of alcohol reaches about _____.

(1) 7% (2) 13% (3) 12% (4) 20%

149.



(i) Label A, B, C and D

(ii) How many CO_2 evolve between A and Malic acid ?

Choose the correct option from the following regarding answers of above questions :-

(1) **(i) A** = PGA, **B** = CO_2, **C** = PGAL, **D** = ATP **(ii)** $3CO_2$

(2) **(i) A** = Citric acid, **B** = CO_2, **C** = Succinic acid, **D** = $FADH_2$ **(ii)** $2CO_2$

(3) **(i) A** = Citric acid, **B** = $FADH_2$, **C** = Succinic acid, **D** = $NADH + H^+$ **(ii)** $2CO_2$

(4) **(i) A** = pyruvic acid, **B** = $FADH_2$, **C** = Succinyl CoA, **D** = GTP. **(ii)** $4CO_2$

150. Select false statement?

(1) Ectomycorrhizal association is found in the roots of *Pinus*.

(2) BGA like- *Nostoc* and *Anabaena* are found in the coralloid roots of *Cycas*.

(3) In conifers, the needle like leaves reduce the surface area. It is xerophytic adaptation.

(4) In Gymnosperm, ovules are covered by ovary wall.

SECTION - A (ZOOLOGY)

151. Consider the following characters :-

1. Bilateral symmetry
2. Acoelomate
3. Incomplete digestive system
4. Mostly endoparasites

Select the correct group of phylum with its corresponding animal for above given characters :-

(1) Arthropoda - cockroach

(2) Ctenophora - Ctenoplana

(3) Aschelminthes - *Taenia solium*

(4) Platyhelminthes - *Fasciola hepatica*

152. **Assertion :-** Cyclostomates are marine but migrate for spawning to fresh water.

Reason :- After spawning, within a few days they die. Their larvae after metamorphosis return to the ocean.

(1) Both Assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason is true but reason is not correct explanation of Assertion.

(3) Assertion is true but reason is wrong.

(4) Assertion and reason both are wrong.

153. In which phylum body is unsegmented with distinct head, muscular foot and visceral hump ?

(1) Mollusca (2) Echinodermata

(3) Hemichordata (4) Annelida

154. On an average, female Periplaneta produces_____oothecae, each containing _____ eggs.

Choose the correct option which fills the blanks respectively.

(1) 14-16, 9-10 (2) 11-12, 17-18

(3) 10-15, 16-20 (4) 9-10, 14-16

155. Fore brain of frog does not include which of the following structures ?

(1) Olfactory lobe (2) Optic lobe

(3) Cerebral hemisphere (4) Diencephalon

156. The most abundant and widely distributed tissue in the bodies of complex animal is

(1) Epithelium (2) Connective

(3) Muscular (4) Neural

157. Match the column-I and column-II.

	Column-I		Column-II
a.	Joint diastole	(i)	0.8 second
b.	Heart beat	(ii)	All four chambers of heart are relaxed.
c.	Cardiac cycle	(iii)	72 times/min
d.	Stroke volume	(iv)	70 mL

Choose the correct answer about given column :-

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(ii), b-(iii), c-(i), d-(iv)

(3) a-(iii), b-(ii), c-(i), d-(iv)

(4) a-(iv), b-(iii), c-(ii), d-(i)

158. Which WBCs are phagocytes cells which destroy foreign organism entering in body ?

(1) Acidophils and basophils

(2) Basophils and monocytes

(3) Neutrophils and monocytes

(4) Neutrophils and lymphocytes

159. Identify enzymes A and B.

$$CO_2 + H_2O \underset{}{\overset{A}{\rightleftharpoons}} H_2CO_3 \overset{B}{\rightleftharpoons} HCO_3^- + H^+$$

(1) A - Carbonic anhydrase, B - Oxidase

(2) A - Oxidase, B - Carbonic anhydrase

(3) A - Carbonic anhydrase, B - Carbonic anhydrase

(4) A - Oxidase, B - Oxidase

160. **Statement-I :** The anatomical set up of lungs in thorax in such that any change in the volume of thoracic cavity will be reflected in the lung cavity.
Statement-II : Volume of air leaving in the lungs even after a forcible expiration is called residual volume.

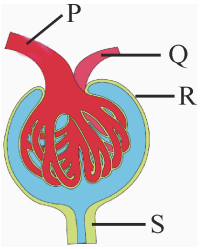
(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

161. In the given diagram, choose the **correct** for labelled parts :-



(1) P ⟶ is fine branch of renal artery

(2) Q ⟶ Carry glomerular filtrate from glomerulus

(3) R ⟶ Glomerulus

(4) S ⟶ Loop of Henle

162. The maximum reabsorption of glomerular filterate occurs in :-

(1) PCT (2) LOH (3) DCT (4) CD

163. Brain stem forms the connections between the ..... and .....

(1) Medulla and pons (2) Brain, spinal cord

(3) Brain, medulla (4) Pons, Spinal Cord

164. Select the correct statement from following.

(1) Oxytocin regulates the growth of the mammary gland.

(2) ACTH stimulates the synthesis and secretion of steroid hormones mineralocorticoid from adrenal cortex.

(3) Vasopressin stimulates reabsorption of water and electrolytes by the distal tubules.

(4) Parathyroid hormone decreases the Ca^{+2} levels in the blood.

165. Deaf mutism can be caused due to :-

(1) Hypersecretion of thymosin

(2) Hyposecretion of thyroxine

(3) Hyposecretion of sex hormone

(4) Hypersecretion of sex hormone

166. Scapula is a large triangular flat bone situated in the dorsal part of the thorax between

(1) The second and fifth ribs

(2) The second and seventh ribs

(3) The third and sixth ribs

(4) The third and eighth ribs

167. The central part of thick filament which is not overlapped by thin filaments is called as :

(1) Z–zone (2) H–zone

(3) I–zone (4) F–zone

168. Consider the following four statements (A–D) and select the option which includes all the correct ones only.

(A) The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens.

(B) Leydig cells synthesise and secrete testicular hormones called androgen.

(C) The female reproductive system is located in the pelvic region.

(D) Each testis has about 250 compartments called testicular lobules.

Options -

(1) A, B, C, D (2) B, C and D Only

(3) A, B and C Only (4) A and B

169. **Statement-I :-** The trophoblast layer gets attached to the endometrium and the inner cell mass gets differentiated as the embryo.

Statement-II :- The morula becomes embedded in the endometrium of the uterus.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

170. Placenta acts as endocrine tissue and produces several hormones. Following are hormones produced in women only during pregnancy except :-

(1) hCG (2) hPL (3) hCS (4) Estrogen

171. The first sign of growing foetus may be noticed by:-

(1) Movement of fetus

(2) Appearance of hair on head

(3) Listening to the heart sound carefully through the stethoscope

(4) Formation of limbs

172. Which of the following has been used for emergency contraception :

(1) Diaphragm (2) Daily tablets

(3) Saheli (4) Unwanted-72

173. Fill in the blanks -

Our body when it encounter a pathogen for the first time produce response called __a__ response which is __b__ intensity. Subsequent encounter with same pathogen elict a __c__ intensified __d__ response.

(1) a-Anamnestic b-Low
c-High d-Secondary

(2) a-Secondary b-High
c-Low d-Primary

(3) a-Primary b-Low
c-High d-Anamnestic

(4) a-Primary b-High
c-Low d-Anamnestic

174. **Assertion (A) :-** In Pneumonia and common cold close contact with infected person should be avoided.

Reason (R) :- Pneumonia and common cold are air-borne diseases.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) Both (A) and (R) are incorrect

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

175. Where immature lymphocytes differentiate into antigen sensitive lymphocyte ?

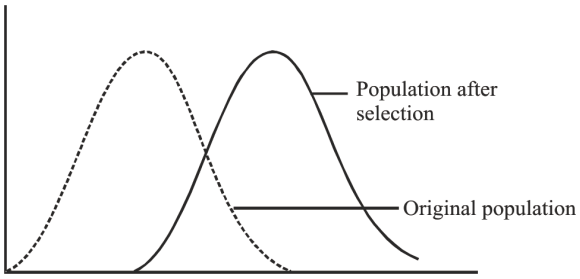
(1) M.A.L.T

(2) Primary lymphoid organs

(3) Secondary lymphoid organs

(4) Spleen

176. Identify the type of natural selection shown in the below mentioned diagrammatic representation :-



(1) Stabilizing selection

(2) Directional selection

(3) Cyclic selection

(4) Disruptive selection

177. The organs which have common origin but perform different functions are called as:

(1) Analogous organs

(2) Homologous organs

(3) Atavistic organs

(4) Vestigial organs

178. Read the following statements carefully :
(A) Lipid component of the plasma membrane mainly consists of phosphoglycerides.
(B) Polar molecules can pass through the lipid bilayer of plasma membrane, therefore they do not require carrier proteins to facilitate their transport.
(C) Secondary wall is capable of growth and it is formed on the outer side of the cell.
(D) Quasifluid nature of lipid enables lateral movement of proteins within the overall lipid bilayer of plasma membrane.
(E) Middle lamella glues the different neighbouring cells together.
How many statements are incorrect ?

(1) Three (2) Five (3) Four (4) Two

179. Process by which ATP is synthesised in cell is named as phosphorylation which organelles are involved in the process?

(1) Leucoplast

(2) Chloroplast only

(3) Chloroplast and mitochondria

(4) Mitochondria and All plastids

180. When a cell is to divide, it synthesizes the various components including the genetic materials during interphase. Choose the correct option for synthesis of materials during interphase ?

(1) All types of DNAs of cell (nuclear and cytoplasmic) synthesis durig S-phase only

(2) Cytoplasmic growth occurs during G_1-phase only.

(3) Protein synthesis occurs during the all sub phases of interphase

(4) All the events of interphase are not genetically controlled

181. The manner in which individual folded polypeptides or subunits are arranged with respect ot each other is :-

(1) Quaternary structure

(2) Tertiary structure

(3) Secondary structure

(4) Primary structure

182. Which one is not a restriction enzyme ?

(1) EcoRI

(2) Pectinase

(3) BamHI

(4) Hind - II

183. Match the following :-

Column-I		Column-II	
(A)	YAC	(I)	Separation of DNA fragments
(B)	Hind II	(II)	Yeast
(C)	Gel electrophoresis	(III)	First restriction endonuclease
(D)	Heterologous host	(IV)	Recombinant protein

(1) A-II, B-III, C-IV, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-III, C-II, D-I

184. Which of the following is not the use of transgenic animal ?

(1) Study of normal physiology and development

(2) Testing of vaccine safety

(3) Production of biological products

(4) Molecular diagnosis

185. Which one is not a biofertilizer ?

(1) *Nostoc*

(2) *Mycorrhiza*

(3) *Agrobacterium*

(4) *Rhizobium*

SECTION - B (ZOOLOGY)

186. 'Chiasmata' is formed between a cross-shaped configuration of :-

(1) sister chromatids of homologous chromosomes.

(2) sister chromatids of non-homologous chromosomes.

(3) non-sister chromatids of homologous chromosomes.

(4) non-sister chromatids of non-homologous chromosomes.

187. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

(a) Lipids are generally water insoluble.

(b) The R group could be a methyl ($-CH_3$) or ethyl ($-C_2H_5$) or higher no. of $-CH_2$ group (1 carbon to 19 carbon) in fatty acids

(c) Palmitic acid has 16 carbons including the carboxyl carbon.

(d) Arachidonic acid has 21 carbon atom including the carboxyl carbon.

Options :-

(1) Statement (a), (b) and (c)

(2) Statement (a), (b) only

(3) Statement (c), (d) only

(4) Statement (a), (c) and (d)

188. The enzyme used during PCR is–

I. Taq polymerase

II. Helicase

III. RNA polymerase

IV. Gyrase

Option :

(1) I only (2) I and III

(3) I and IV (4) I, II and III

189. Which is true for *Spongilla*?

(1) Occurs exclusively in fresh water

(2) Is a triploblastic poriferan

(3) Has numerous osculum for entry of water

(4) Lacks flagellated collar cells in body

190. Largest phylum, jointed appendages, chitinous exoskeleton, mostly dioecious The above features belongs to :-

(1) Pheretima (2) Fasciola

(3) Apis (4) Ascaris

191. Coronary artery disease often referred as :-

(1) Angina (2) Heart failure

(3) Cardiac arrest (4) Atherosclerosis

192. Match the column-I with column-II

Column-I		Column-II	
A	Alveoli	i	0.3 mmHg PCO_2
B	Deoxygenated blood	ii	45 mmHg PCO_2
C	Atmospheric air	iii	104 mmHg PO_2
D	Tissue	iv	40 mmHg PO_2

(1) A - iii, B - iv, C - ii, D - i

(2) A - iii, B - iv, C - i, D - ii

(3) A - iv, B - iii, C - i, D - ii

(4) A - ii, B - i, C - iii, D - iv

193. NaCl from ascending Henle's loop is exchanged with

(1) ascending limb of vasa recta

(2) descending limb of vasa recta

(3) afferent arteriole

(4) descending limb of Henle's loop

194. Which of the following is not a part of thin filament?

(1) Troponin (2) Tropomyosin

(3) Actin (4) Meromyosin

195. Choose the correct option :-
(a) First seven pairs of ribs are called true ribs.
(b) Sternum is a large triangular bone that situated at the dorsol part of the thorax.
(c) Scapula has a slightly elevated ridge called the spine which projects as a flat, expanded process called the acromion.
(d) Below the acromion is a depression called the glenoid cavity which articulates with the head of the femur.

(1) only a, b and c (2) only a and c

(3) only b, c and d (4) Only a

196. How many primary follicles are left in each ovary in human female at puberty?

(1) 200 - 300 million (2) 15 - 20

(3) 120000 - 1,60,000 (4) 60,000 - 80,000

197. MTP is performed to get rid of unwanted pregnancies due to :-

(1) Failure of the contraceptive used during coitus

(2) Rape

(3) Casual unprotected intercourse

(4) All of these

198. **Assertion A :** Colostrum provides passive immunity to the infant.
Reason R : Colostrum secreted by the mother contains antibodies that protect the infant.

(1) **A** is correct but **R** is not correct.

(2) **A** is not correct but **R** is correct.

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.

199. Which is not correctly matched among the following options ?

(1) Homo habilis - Brain capacity 650 - 800 cc

(2) Homo erectus - Fossils discovered in Java

(3) Neanderthal man - First human being like creature

(4) Homo sapiens - Arose in Africa

200. Which of the following epithelium is not present in nephron ?

(1) Squamous epithelium

(2) Brush bordered cuboidal epithelium

(3) Ciliated columnar epithelium

(4) Cuboidal epithelium